

CS 510: Randomized Algorithms

Course Code	CS 510
Course Name	Randomized Algorithms
L-T-P-C	4-0-0-4
Prerequisites	Data Structures and Algorithms, Basic Discrete Probability
Intended for	M.Tech/Ph.D./B.Tech 3rd/4th year
Approval	57th BoA (Oct 2024)

Course Contents

- **Introduction:** review of discrete probability, Introduction to randomized algorithms, basic examples: Polynomial Identity Testing, matrix product verification (Frievald's algorithm), Karger's randomized min-cut algorithm, randomized quicksort. [8 hours]
- **Concentration inequalities:** Markov's and Chebychev's inequalities, randomized selection, two-point sampling, stable marriage problem, coupon collector's problem, birthday paradox, balls and bins, Chernoff bound, application to error reduction and network routing. [8 hours]
- **Markov chains and random walks:** randomized algorithm for 2SAT, Markov chains, random walks on graphs, graph connectivity, expander graphs and randomly mixing random walks, probability amplification using random walks on expander graphs. [9 hours]
- **Randomized data structures:** random treaps, skip lists, hash tables, universal family of hash functions, perfect hashing. [8 hours]
- **Randomized computational geometry:** randomized incremental construction, convex hull construction, geometric duality, half space intersections, Delaunay triangulation, trapezoidal decomposition, random sampling, linear programming. [8 hours]
- **Randomized graph algorithms:** all pairs shortest paths, minimum-cut, minimum spanning trees. [7 hours]
- **Online algorithms:** adversary models, paging against oblivious and adaptive adversaries, the k-server problem. [6 hours]
- **Randomized Complexity classes:** RP, co-RP, ZPP, BPP. [2 hours]

Text Books

1. R. Motwani and P. Raghavan, **Randomized Algorithms**, Cambridge University Press, 1995.
2. M. Mitzenmacher and E. Upfal, **Probability and Computing: Randomization and probabilistic techniques in algorithms and data analysis**, 2nd Edition, Cambridge University Press, 2017.
3. S. P. Vadhan, **Pseudorandomness**, Now Publishers, 2012.

References

1. D. P. Dubhashi and A. Panconesi, **Concentration of Measure for the Analysis of Randomized Algorithms**, Cambridge University Press, 2009.